7.7 Quantities That Never Change

WORKSHEET

The question

STEP III 2024, Question 5

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The expression $\mathbf{MN} - \mathbf{NM}$ is called the *commutator* of $\mathbf{M}$ and $\mathbf{N}$. When is it zero? What is its trace, given part (i)?

STAGE 1 The trace

Covers: Part (i)

Ask yourself Write $\mathbf{N} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ and work out only the diagonal entries of $\mathbf{MN}$ and of $\mathbf{NM}$.

STAGE 2 The derivative of a determinant

Covers: Part (ii)

Ask yourself Differentiate $ad - bc$ by the product rule. Then work out the diagonal entries of $\mathbf{M}^{-1}\frac{d\mathbf{M}}{dt}$, using the formula for the inverse of a 2×2 matrix.

STAGE 3 Three conserved quantities

Covers: Part (iii), the three constants

Ask yourself For the trace, take the trace of both sides of the differential equation and use Part (i).

Pause and try For the determinant, multiply the equation on the left by $\mathbf{M}^{-1}$ and use Part (ii). You will need $\operatorname{tr}(\mathbf{M}^{-1}\mathbf{N}\mathbf{M}) = \operatorname{tr}\mathbf{N}$: why is that true? For $\operatorname{tr}(\mathbf{M}^2)$, write it in terms of $\operatorname{tr}\mathbf{M}$ and $\det\mathbf{M}$.

STAGE 4 An explicit solution

Covers: Part (iii), solving for $\mathbf{M}$

Ask yourself Split $\mathbf{N}$ into $t\mathbf{I}$ plus $t\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix}$. The $t\mathbf{I}$ part commutes with $\mathbf{M}$, so it drops out of $\mathbf{MN} - \mathbf{NM}$.

Pause and try Write the four differential equations for a, b, c, d. Solve for c first, then a and d, then b. Check that your trace and determinant are constant.

STAGE 5 A different equation

Covers: Part (iv)

Pause and try If $\frac{d\mathbf{M}}{dt} = \mathbf{MN}$, then $\mathbf{N} = \mathbf{M}^{-1}\frac{d\mathbf{M}}{dt}$, and Part (ii) links $\operatorname{tr}\mathbf{N}$ to how $\det\mathbf{M}$ changes. Look for an $\mathbf{M}$ whose trace stays fixed while its determinant changes, such as $\begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}$.

What's really going on

Solitons and Lax pairs

In 1834 the Scottish engineer John Scott Russell followed, on horseback, a solitary wave travelling along the Union Canal near Edinburgh for over a mile without losing its shape. Such waves are described by the *Korteweg–de Vries (KdV) equation*, and in the 1960s computer experiments by Norman Zabusky and Martin Kruskal showed that two of these "solitons" can collide and emerge unchanged, as if they were particles. Explaining this needed infinitely many conserved quantities. Peter Lax showed in 1968 that the KdV equation can be written as $\dot{L} = LP - PL$ for a pair of differential operators L and P; then, exactly as in this question, the eigenvalues of L are conserved, and they provide the infinitely many conservation laws. Systems with a Lax pair are called *integrable*, and they are among the few nonlinear systems that can be solved exactly.

Isospectral flows

Since the trace and determinant of $\mathbf{M}$ determine its characteristic polynomial $\lambda^2 - (\operatorname{tr}\mathbf{M})\lambda + \det\mathbf{M}$, the eigenvalues of $\mathbf{M}$ do not change with time. The flow $\dot{\mathbf{M}} = [\mathbf{M}, \mathbf{N}]$ moves $\mathbf{M}$ around inside the set of matrices with fixed eigenvalues. In fact, the solution is always of the form $\mathbf{M}(t) = \mathbf{S}(t)^{-1}\mathbf{M}(0)\mathbf{S}(t)$ for some invertible $\mathbf{S}(t)$: the matrix stays similar to its starting value. The same idea runs through physics: in quantum mechanics, an observable $\mathbf{A}$ evolves by $\frac{d\mathbf{A}}{dt} = \frac{i}{\hbar}[\mathbf{H}, \mathbf{A}]$, where $\mathbf{H}$ is the energy, so anything that commutes with the energy is conserved.

Jacobi's formula

Part (ii) holds for invertible matrices of any size and is called *Jacobi's formula*. It is safest written without the logarithm, as $\frac{\mathrm{d}}{\mathrm{d}t}\det\mathbf{M} = \det\mathbf{M}\,\operatorname{tr}\!\left(\mathbf{M}^{-1}\dot{\mathbf{M}}\right)$, because $\det\mathbf{M}$ may be negative; the logarithmic form $\frac{\mathrm{d}}{\mathrm{d}t}\ln\det\mathbf{M} = \operatorname{tr}\!\left(\mathbf{M}^{-1}\dot{\mathbf{M}}\right)$ needs $\det\mathbf{M} > 0$ (or $\ln|\det\mathbf{M}|$ in general). A consequence: if $\dot{\mathbf{M}} = \mathbf{A}\mathbf{M}$ for a constant matrix $\mathbf{A}$, then $\det\mathbf{M}(t) = e^{t\operatorname{tr}\mathbf{A}}\det\mathbf{M}(0)$. In the study of differential equations this tells you how volumes in phase space grow or shrink; for the equations of classical mechanics the trace is zero and volume is preserved, a result known as *Liouville's theorem*.

Try this next

1. **Check the solution.** Verify by differentiating that the matrix $\mathbf{M}(t)$ in Stage 4 satisfies $\dot{\mathbf{M}} = \mathbf{M}\mathbf{N} - \mathbf{N}\mathbf{M}$.

2. **Similarity.** Show that if $\mathbf{M}(t) = \mathbf{S}^{-1}\mathbf{M}_0\mathbf{S}$, where $\mathbf{M}_0$ is constant and $\mathbf{S}(t)$ satisfies $\dot{\mathbf{S}} = \mathbf{S}\mathbf{N}$, then $\dot{\mathbf{M}} = \mathbf{M}\mathbf{N} - \mathbf{N}\mathbf{M}$. (*Hint:* $\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{S}^{-1} = -\mathbf{S}^{-1}\dot{\mathbf{S}}\mathbf{S}^{-1}$.)

3. **Volume growth.** Let $\mathbf{M}(t) = \begin{pmatrix} e^{2t} & 0 \\ 0 & e^{-t} \end{pmatrix}$. Find $\mathbf{N}$ with $\dot{\mathbf{M}} = \mathbf{M}\mathbf{N}$, and check that $\frac{\mathrm{d}}{\mathrm{d}t}\ln\det\mathbf{M} = \operatorname{tr}\mathbf{N}$.

4. **Higher powers.** Show directly that $\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{tr}(\mathbf{M}^k) = 0$ for every positive integer k when $\dot{\mathbf{M}} = \mathbf{M}\mathbf{N} - \mathbf{N}\mathbf{M}$.

Further reading

- **P. G. Drazin & R. S. Johnson, *Solitons: An Introduction*** (Cambridge University Press). The standard undergraduate introduction, including Lax pairs.
- **Thierry Dauxois & Michel Peyrard, *Physics of Solitons*** (Cambridge University Press). Includes the story of Scott Russell's wave.
- **Dwight Neuenschwander, *Emmy Noether's Wonderful Theorem*** (Johns Hopkins University Press). How symmetries give conservation laws.

The full guided solution is in Section 7.7:
https://beyondstep.keithmcnulty.org/ch07/lax-pairs.html